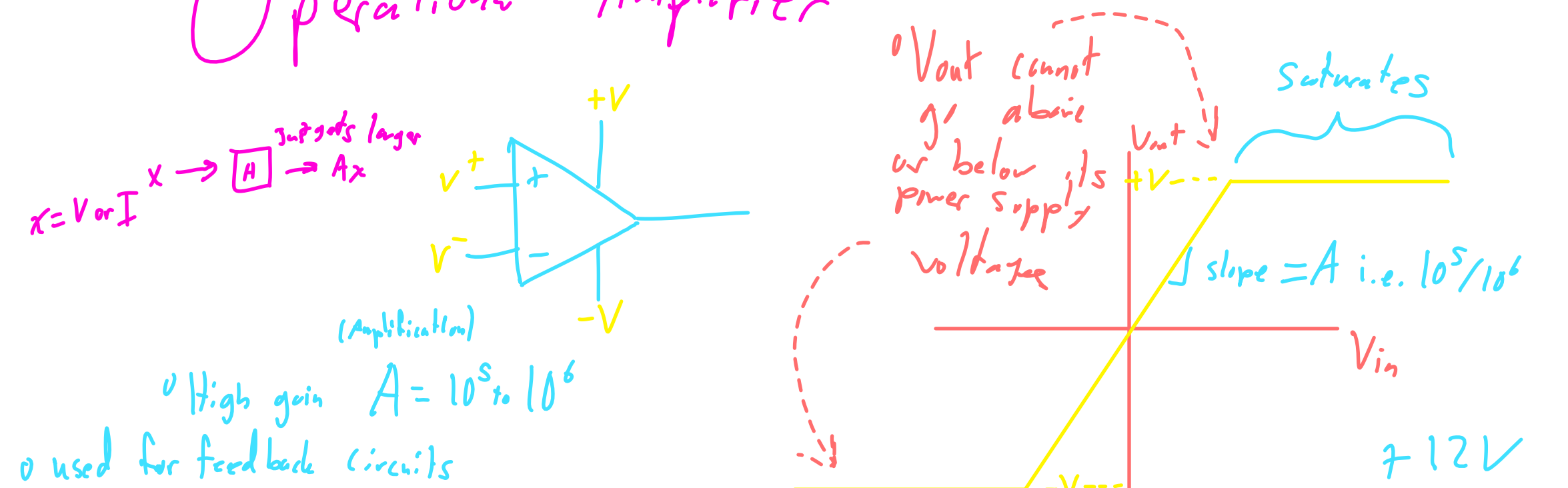
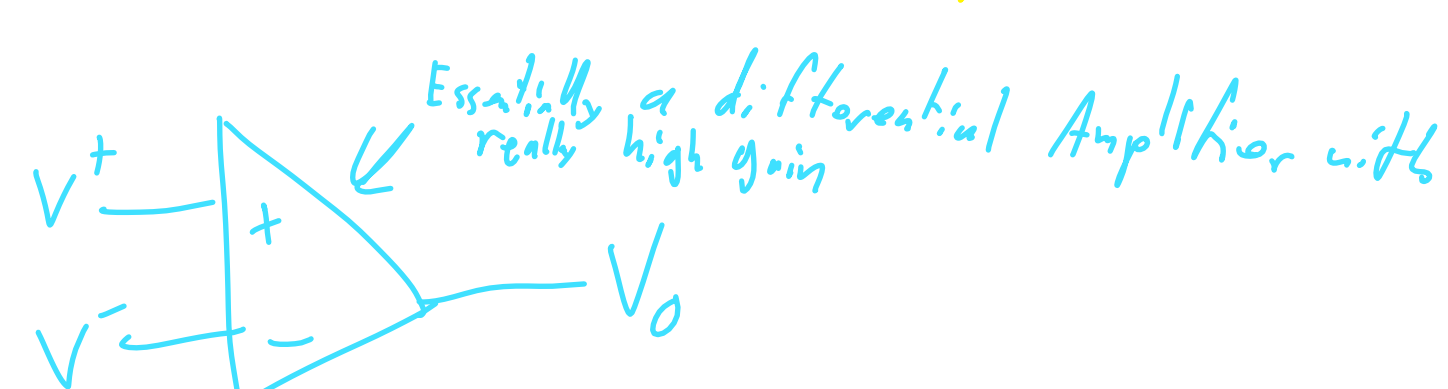


Operational Amplifier



Op Amps usually have Differential inputs (not single-ended inputs, $V_0 = A(V^+ - V^-)$)



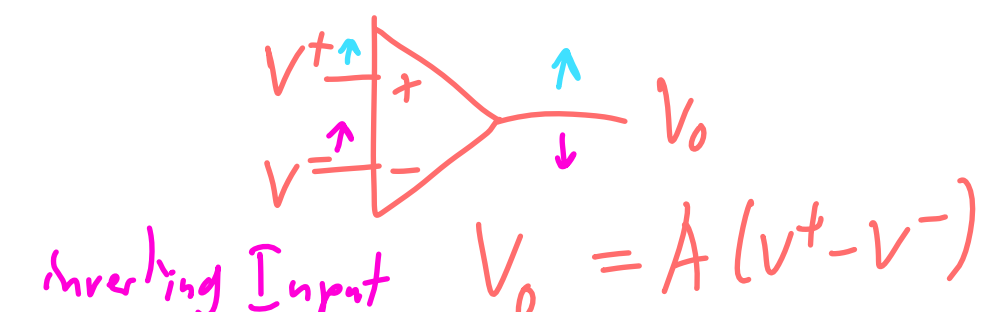
In an ideal op-amp, $I = 0$

$$V_0 = A(V^+ - V^-)$$

$$V_0 = A V_{in}$$

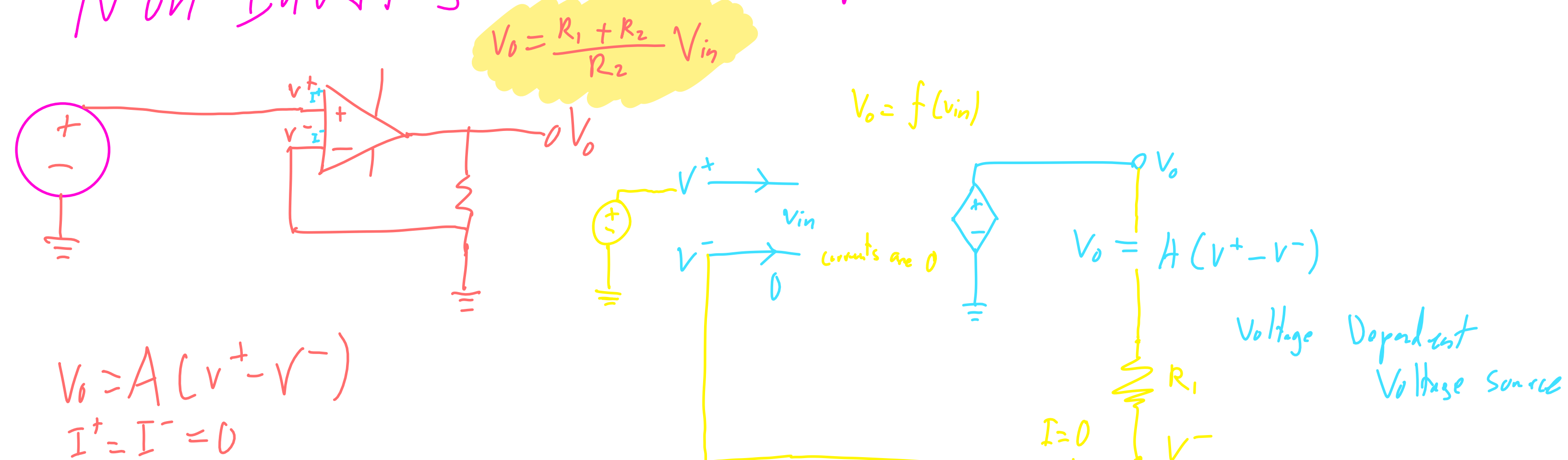
$$V_{in} = V^+ - V^-$$

Non Inverting Input



V_{out} goes up with Non Inverting Inputs and down with Inverting Inputs

Non Inverting Operational Amplifier



$$V_0 = A(V^+ - V^- \frac{R_2}{R_1 + R_2})$$

$$V_0 = A V^+ - A V^- \left(\frac{R_2}{R_1 + R_2} \right)$$

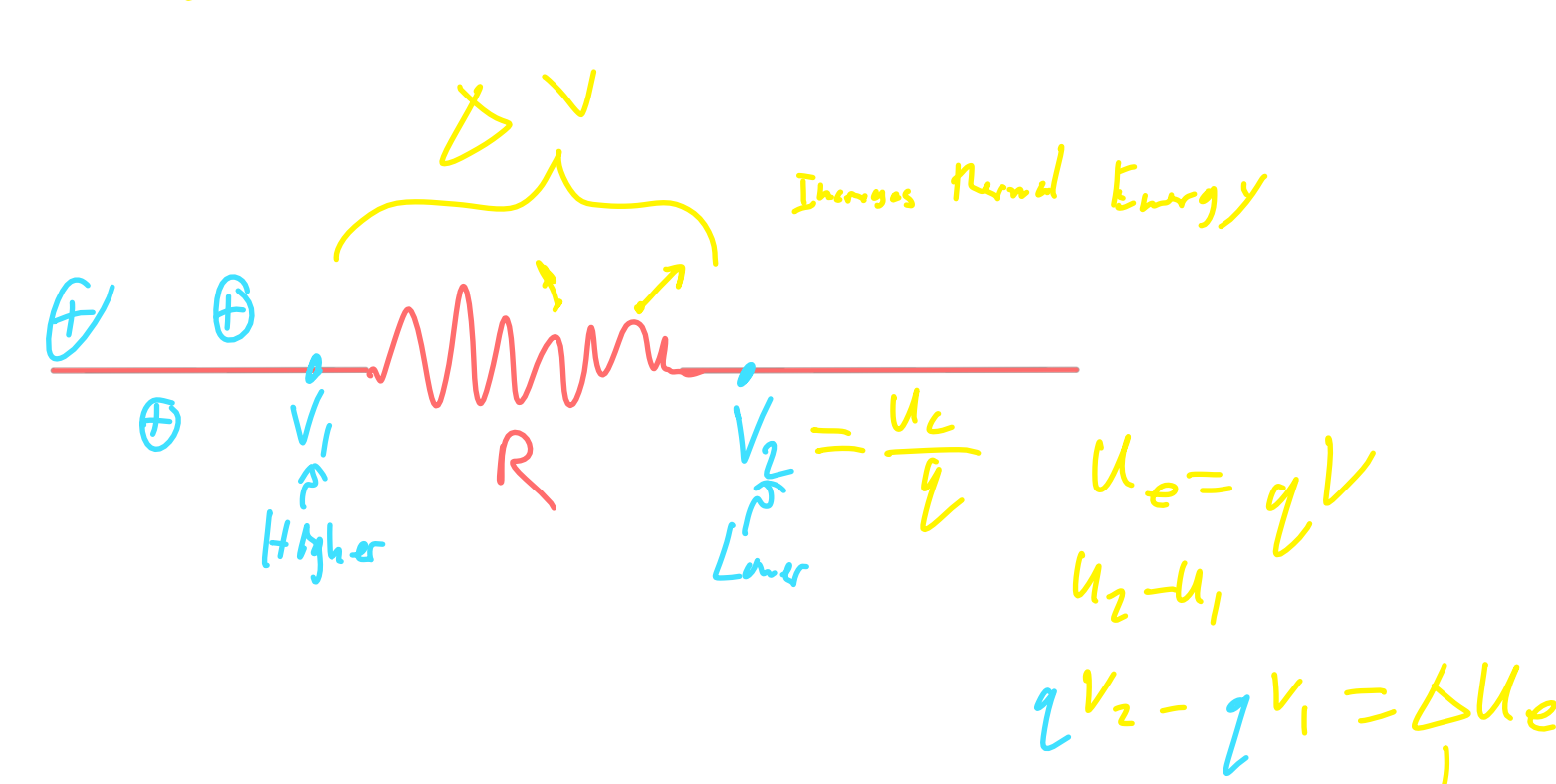
$$V_0 + A V^- \left(\frac{R_2}{R_1 + R_2} \right) = A V_{in}$$

$$V_0 \left(1 + \frac{A R_2}{R_1 + R_2} \right) = A V_{in}$$

$$V_0 = \frac{A V_{in}}{1 + \frac{A R_2}{R_1 + R_2}} \quad \left\{ \begin{array}{l} \text{proportional to } R_1, R_2, \dots \text{ and } A \\ \text{is independent of } A \end{array} \right.$$

$$V_0 = \frac{R_1 + R_2}{R_2} V_{in}$$

Electric Power



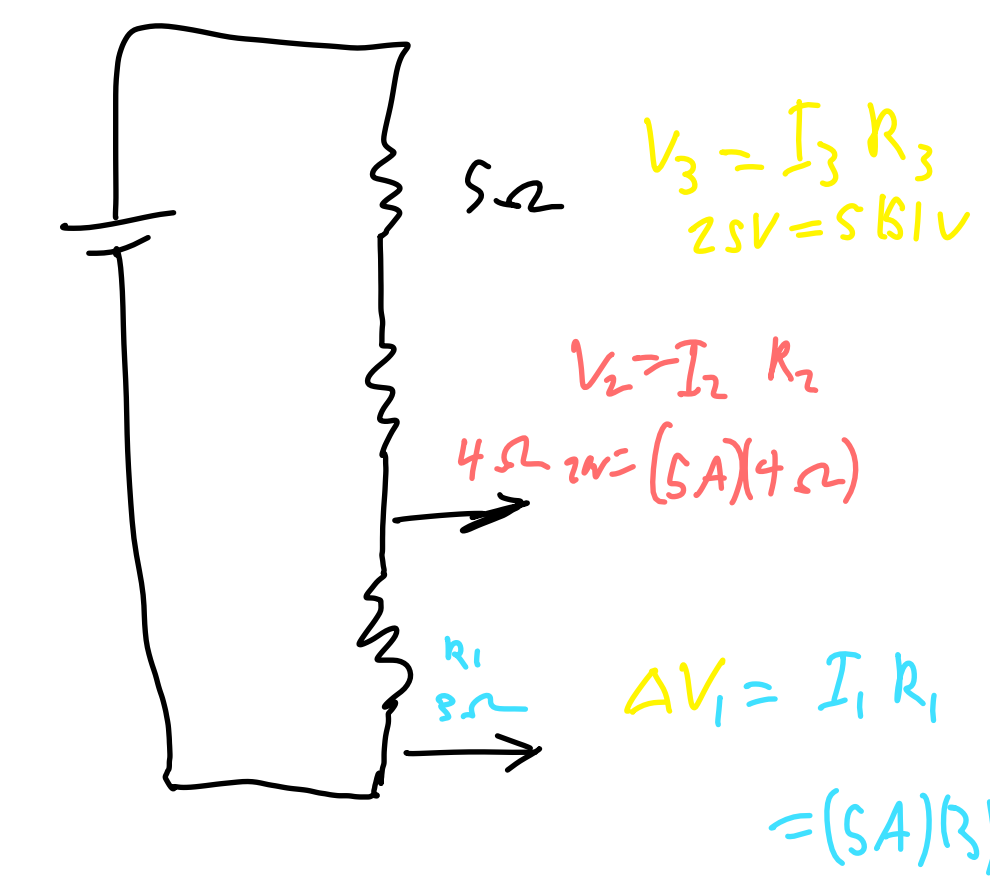
$$P = \frac{\Delta E}{t} = \frac{\Delta U_c}{t}$$

$$P = \frac{e(U_2 - U_1)}{t} = \frac{e \Delta V}{t}$$

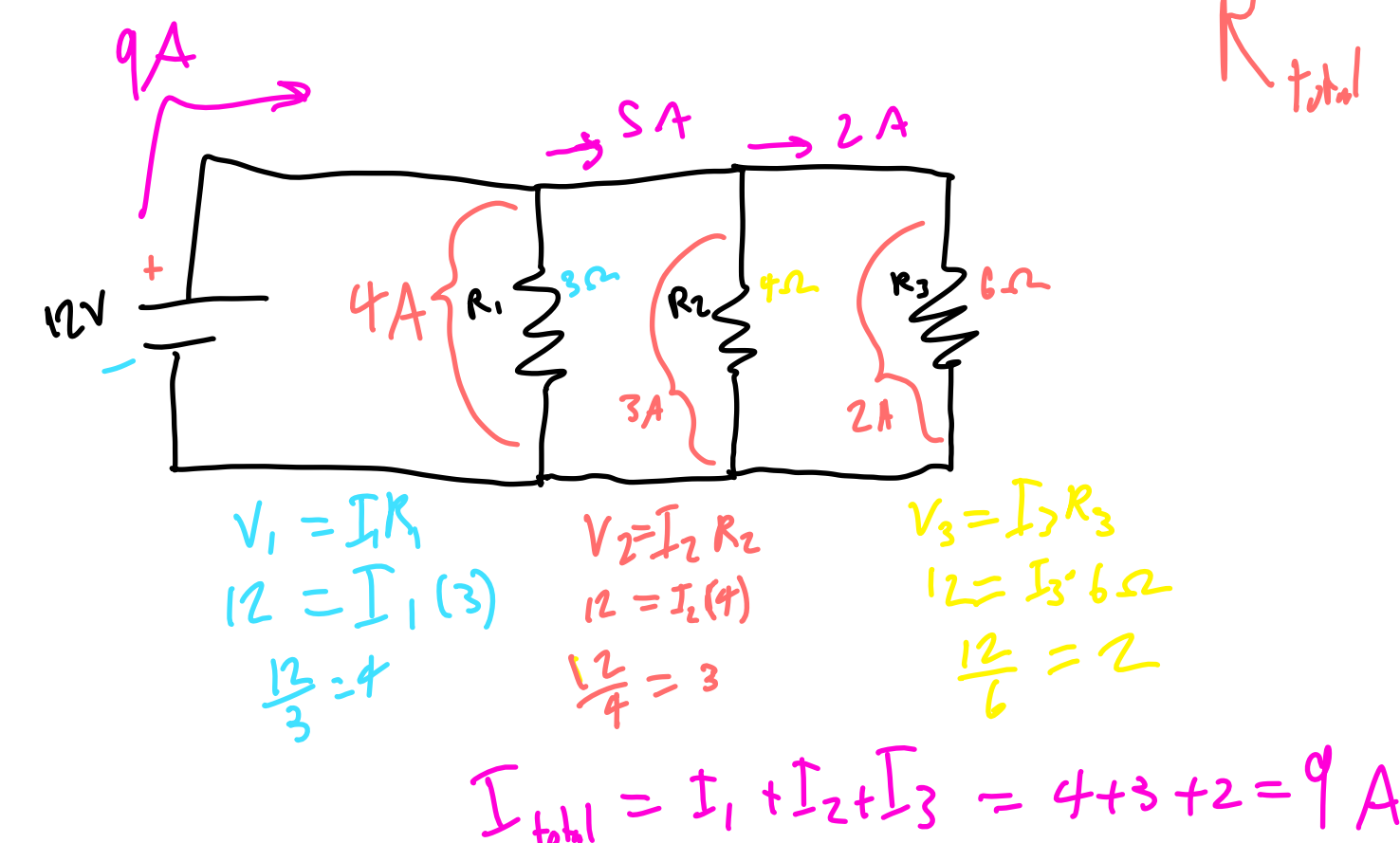
$$I = \frac{q}{t}$$

$$P = IAV$$

Ohm's Law



Kirchoff's Law



$$\frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Feed Back

